

M2 Bio-Informatic Internship

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1 Introduction

2 Results

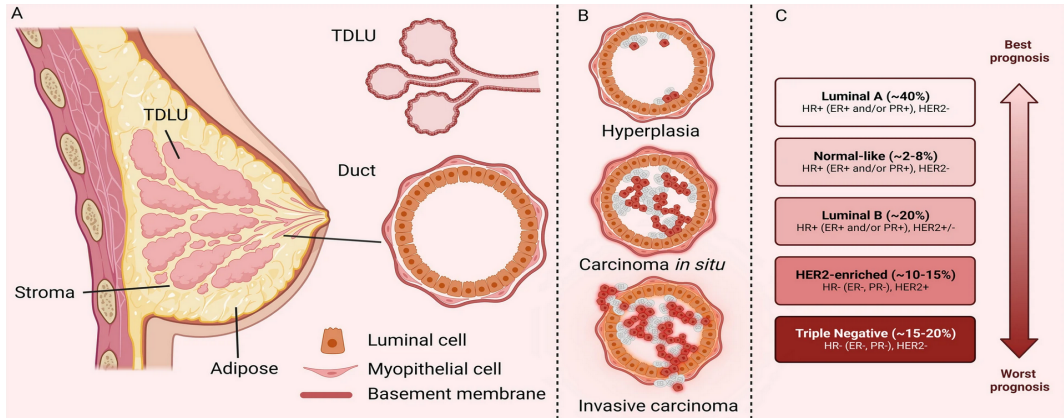
Phenotypic markers analysis

Copy Number Alterations (CNA) analysis

3 Conclusions

4 Future Work

Context

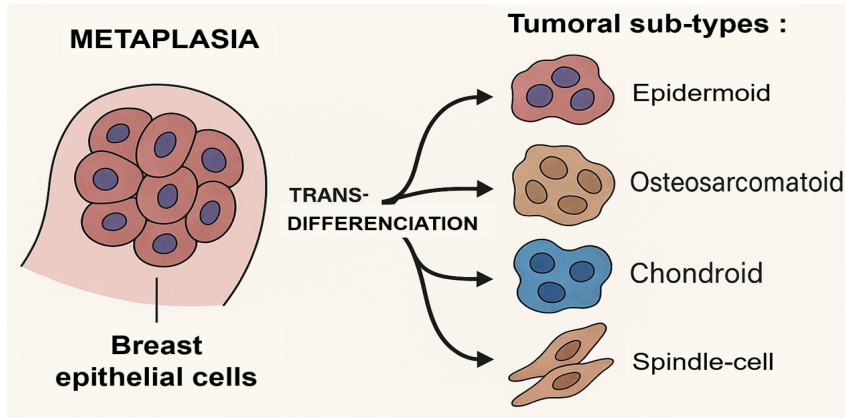


TNBC: Triple-Negative Breast Cancer
TDLU: Terminal Duct Lobular Units

Source: Chang-Yun Li
biomedcentral.com (CC BY 4.0)

MpBC (Metaplastic Breast Carcinomas) are rare forms of TNBC, lacking molecular diagnostic markers and specific therapies

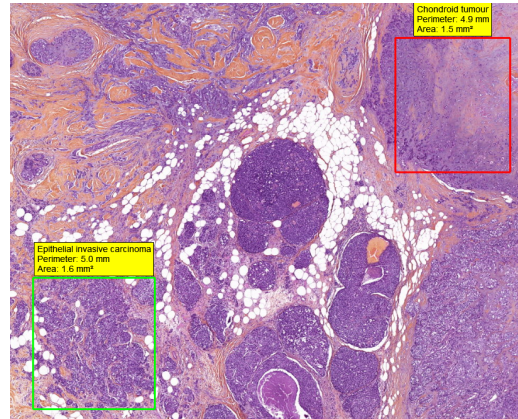
Trans-differentiation



MpBC can transdifferentiate into various aggressive tumor subtypes

Trans-differentiation

- **MpBC** exhibits a remarkable **plasticity**
- **Transdifferentiation** into multiple aggressive **tumor subtypes**
- **Mixed MpBC**
 - Each sample contains **at least 2 tumoral compartments**



Research questions

Unresolved questions

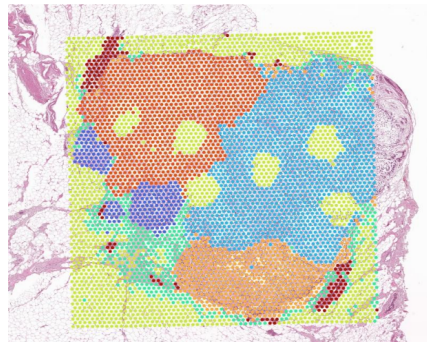
- ① Understanding the **evolutionary trajectories** in MpBC
 - Internal determinants (genetic and epigenetic)
 - External determinants (tumor microenvironment)
- ② Identify molecular biomarkers to **improve diagnostic** precision
- ③ Discover genetic and epigenetic features that may serve as potential **therapeutic targets**

Internship Aims

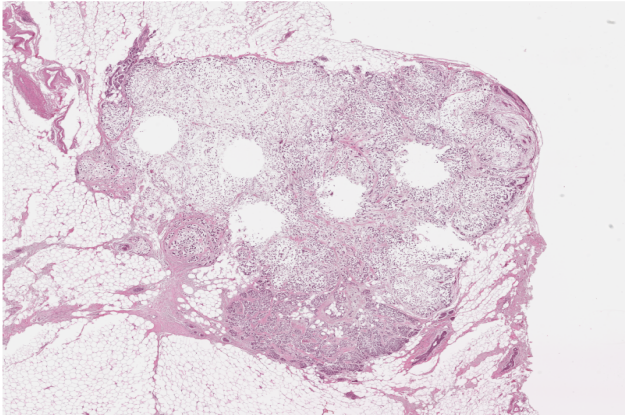
- Determine **expression markers** specific to different tumor sub-types
- Analyze **genomic divergence** between tumoral compartments

Data

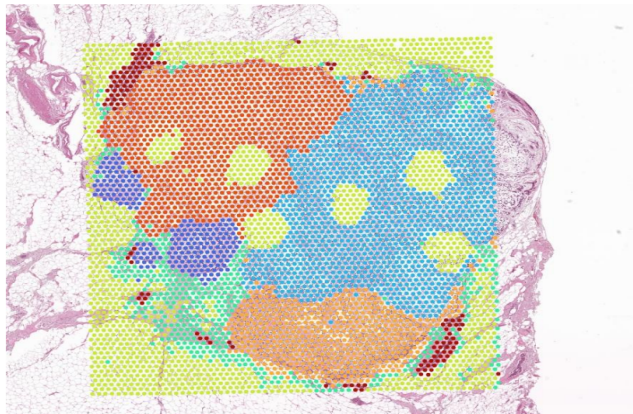
- **Spatial transcriptomic counts** (*Visium*, 10X Genomics)
- **16 mixed MpBC** samples
- Different tumor **transdifferentiation** states captured



MpBC sample



Expert annotations



- Artifacts
- Classical chondrosarcoma cells
- Epithelial tumor cells
- Intermediate tumour cells
- Mixoid chondrosarcoma cells
- Normal fibrous tissue
- Normal epithelium

Mixed MpBC with several tumor cell types covered by Visium spots, grouped by k-means clustering and annotated by a pathologist

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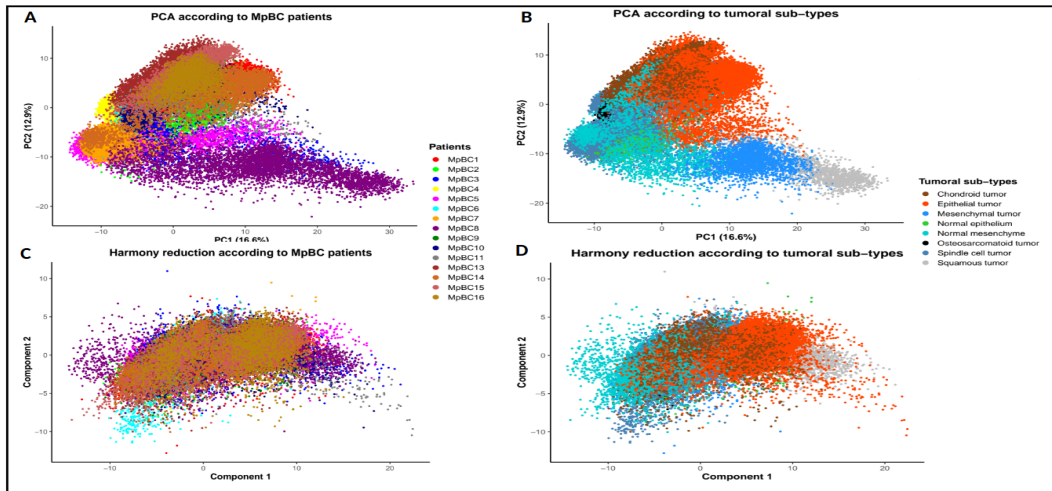
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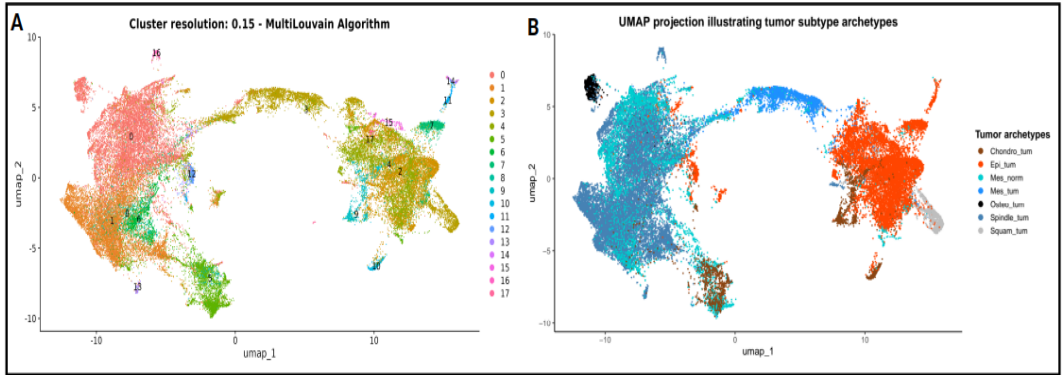
4 Future Work

Batch effect correction : Harmony



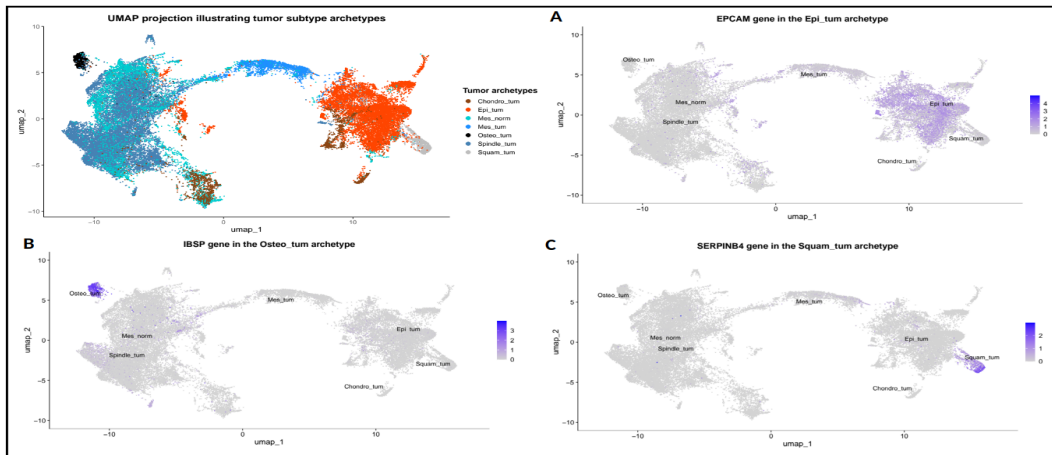
Improve integration with fewer isolated patients

UMAP Projection



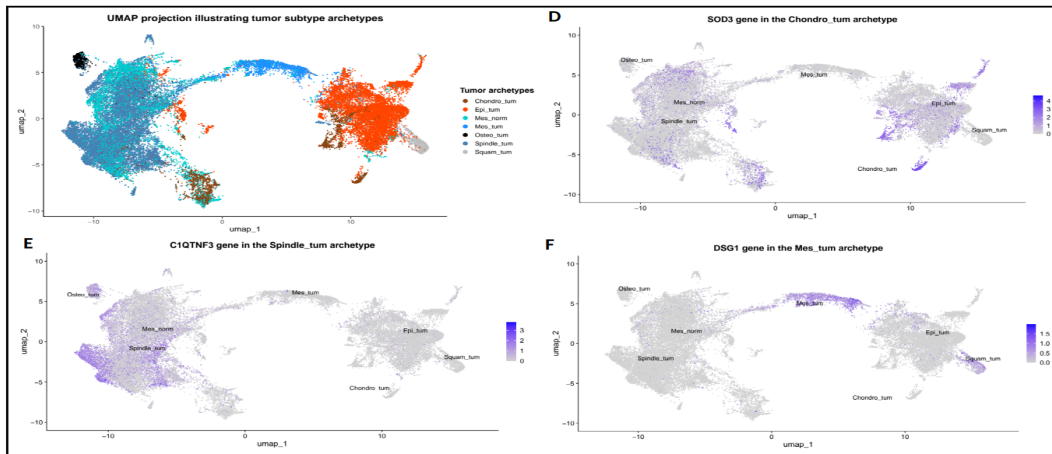
Clear epithelial-mesenchymal axis (umap1), with few ambiguous spots

Expression markers



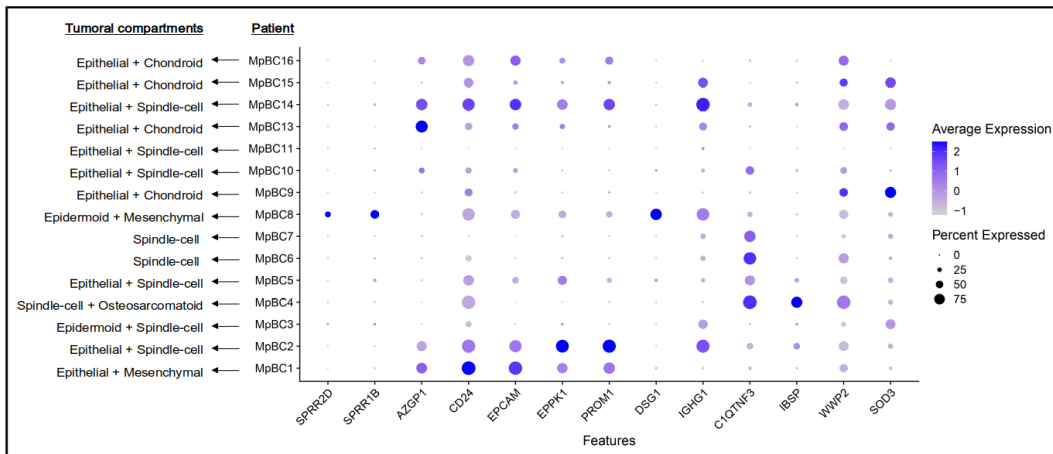
Analysis with **MAST** (Model-based Analysis of Single-cell Transcriptomics)
Some phenotypic markers identified are specific to clusters...

Expression markers



Few phenotypic markers aren't specific enough (Visium limitations)
Different spot purity depending on the subtypes

Markers specificities



Some cluster-specific markers are also patient-specific markers

Analysis conclusions

- Analysis reveals **archetype-specific markers**
 - **Epidermoid**
 - **Epithelial**
 - **Osteosarcomatoid**
- Some markers remain **non-specific or not representative of the archetypes**
 - **Spindle-cell**
 - **Chondroid**
 - **Mesenchymal**
- **Spatial transcriptomics** combined with **pathologist annotations** is **promising**, but **still limited** by the **cellular purity of Visium spots**

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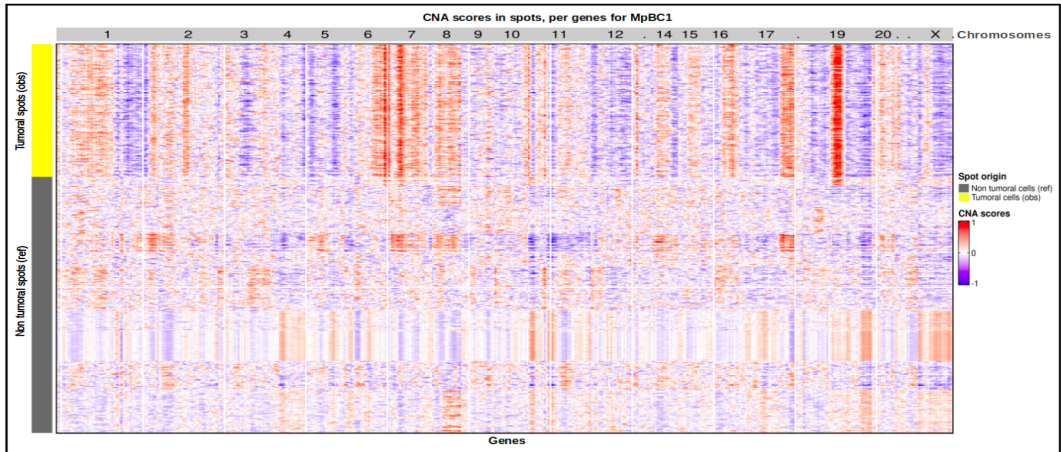
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CNA in cancer

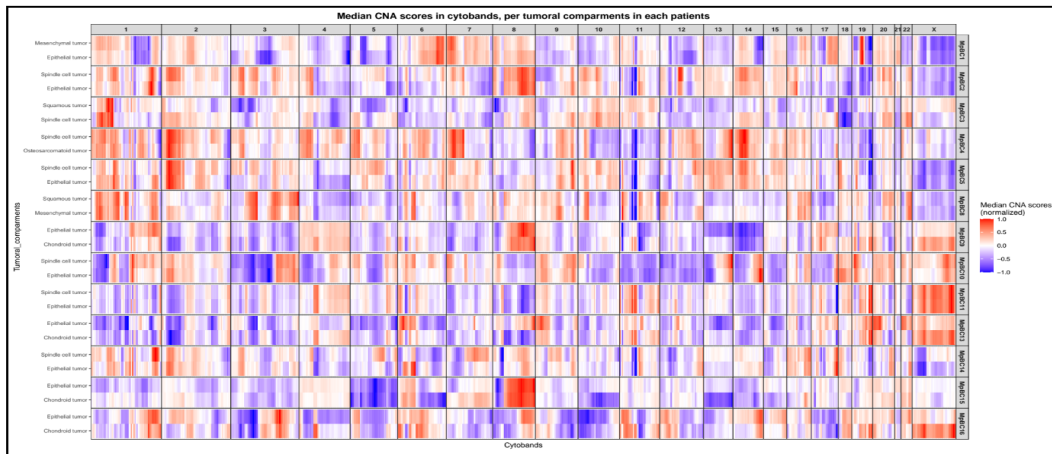
- **Frequent in cancer:** CNA are a hallmark of cancer cells.
- **Types of alterations:**
 - Can be **focal** (targeting specific genes or regions)
 - Or **broad**, affecting entire cytobands or chromosome arms
- **Functional impact:**
 - CNA can lead to oncogene amplification or tumor suppressor loss
- **Clonal evolution insight:**
 - CNA profiles help reconstruct **evolutionary trajectories**
 - Reveal selection of clones and subclones in tumor

Raw InferCNVPlus results



Raw CNA scores in tumoral (obs) or non tumoral (ref) spots for each gene expressed

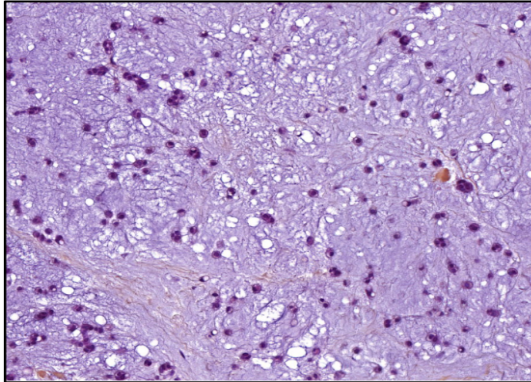
Heatmap



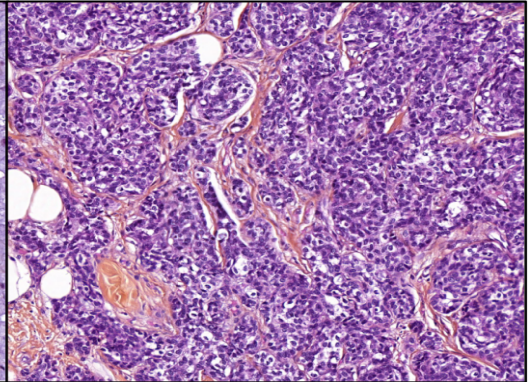
CNA scores in each tumoral compartments for each cytoband

RNA sequencing depth per spot

Chondroid tumoral cells

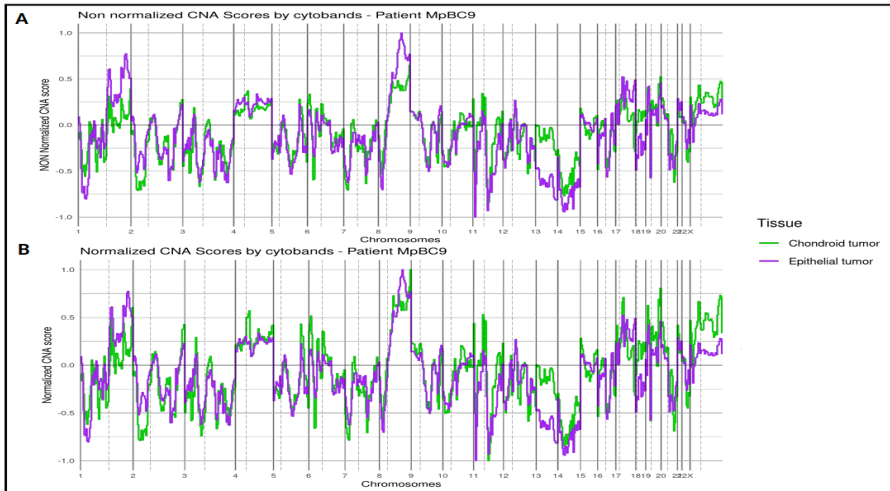


Epithelial tumoral cells



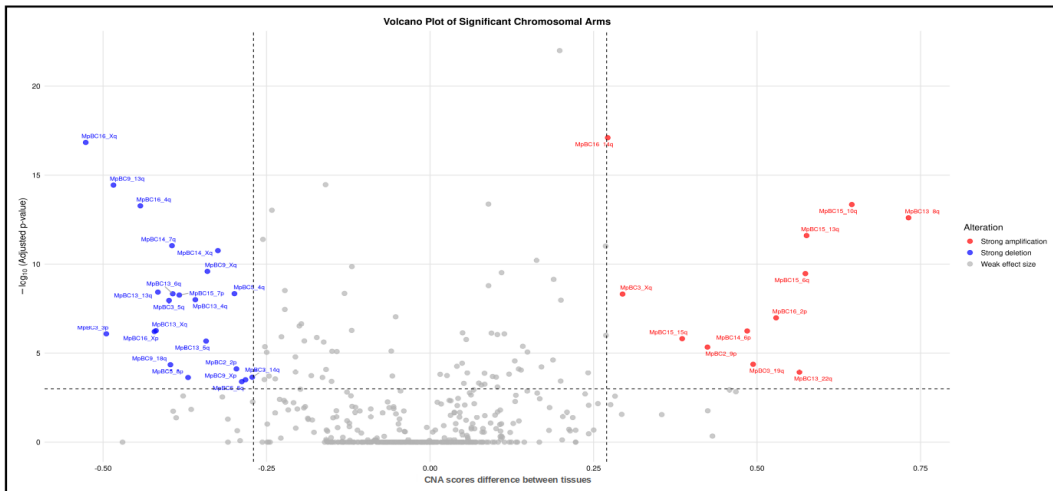
Normalization to correct for differences in RNA sequencing depth across spots

Genomic CNA profile



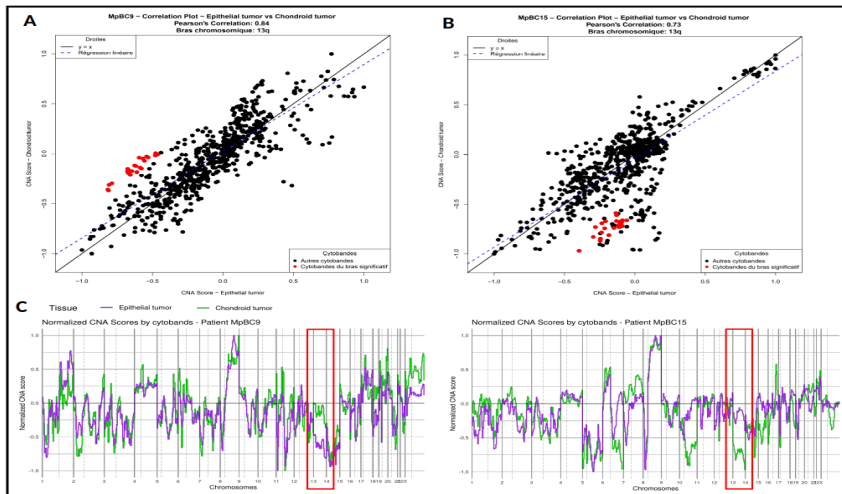
Some genomic regions are highly altered even after the normalisation

VolcanoPlot



Some chromosomal arms significantly altered for few patients

Correlation Plot



Significant cytobands deviate from other cytoband distributions

Analysis Conclusions

- **Limited divergence** in CNA profiles between paired tumor compartments
 - Suggests a **shared common origin** for tumor subtypes
- **Rare cases** of chromosomal arm-level divergence between compartments
 - May reflect **clonal evolution**, where one compartment originates from the other

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Conclusion

Main results

- Identified **phenotypic markers** specific to subtypes, but there is still limits (Visium resolutions)
 - Emphasizes the need for single-cell resolution
- **Limited genomic divergence** suggests a common origin of tumoral compartments in MpBC

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Future Work

1. Perform snRNA-seq on MpBC

- Characterize MpBC subtypes using **more specific molecular markers**
- Build a **MpBC-specific transcriptomic atlas** for each tumor subtype

2. Visium Spot Deconvolution

- Apply **spot deconvolution algorithms** (e.g., RCTD) to estimate cell type composition per spot
- Improve the assignment of transcriptomic profiles to specific tumor subtypes

Future Work

3. Microdissection of Tumoral Compartments

- Perform exome sequencing to explore **intrinsic molecular determinants**
- Detect point mutations, driver alterations, and validate CNA findings
- Conduct epigenetic profiling, including methylome analysis

4. Long-term Objectives: Clinical Applications

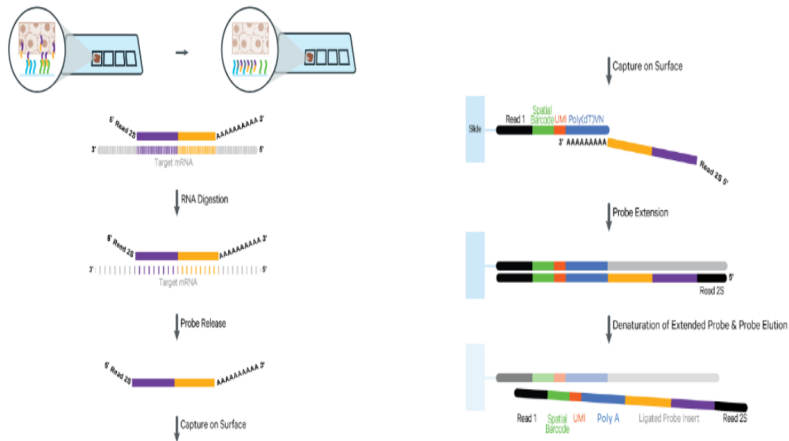
- Discover **subtype-specific molecular markers**
- Identify **targetable pathways** relevant to MpBC

Thank you for listening !

Tutor : Dr Pierre Martinez

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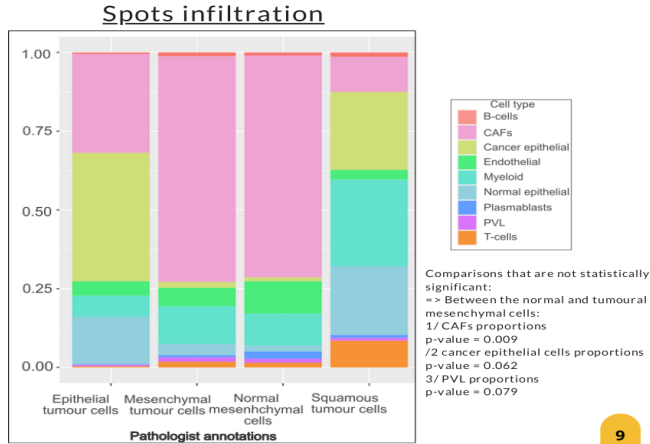
FFPE-Visium workflow



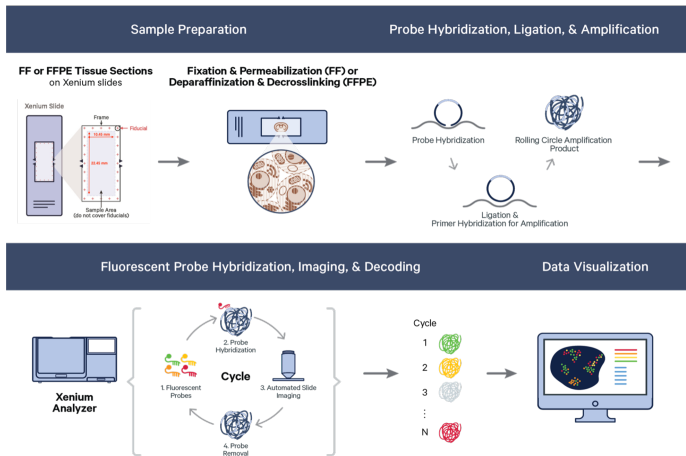
Source: 10x Genomics

Probe hybridization and ligation to capture RNA in each spots

RCTD deconvolution



Xenium



Source: 10x Genomics